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Department of Biology
Northeastern University
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EDUCATION

University of Wisconsin-Madison (UW-Madison)

Madison, WI

Ph.D. in Microbiology (Advisor: Prof. Doug Weibel, Ph.D.)

2011–2017

Universidade Federal de Viçosa (UFV)

Viçosa (Brazil)

M.Sc. in Agricultural Microbiology (Advisor: Prof. Arnaldo Borges, Ph.D.)

2006–2008

Universidade Federal de Viçosa

Viçosa (Brazil)

B.Sc. in Biological Sciences (*summa cum laude*) (Advisor: Prof. Célia Moraes, Ph.D.)

2002–2006

APPOINTMENTS

Northeastern University

Boston, MA

Department of Chemistry & Chemical Biology

Assistant Professor of Chemistry & Chemical Biology

2024–present

Assistant Professor of Biology

2024–present

Harvard University

Cambridge, MA

Department of Chemistry & Chemical Biology

Postdoctoral Researcher (Prof. Daniel Kahne, Ph.D.)

2017–2023

Cornell University

Ithaca, NY

Department of Population Medicine & Diagnostic Sciences

Research Scholar (Prof. Rodrigo Bicalho, D.M.V, Ph.D.)

2008–2011

Publications Since Starting at Northeastern University: 3

Notation: [‡] = corresponding author; ^{*} = undergraduate student advised by TMA; [#] = graduate student or visiting scholar advised by TMA; [†] = postdoctoral researcher of research scholar supervised by TMA; * = equal contribution.

25. Dilmaghani-Marand A[#], Santos LM^{*†}, Sousa TF[†], Santos TMA[‡]. Periplasmic and extracellular regions of the assembly machine protein BamA actively facilitate outer membrane protein folding. (*In preparation for submission, Fall 2026*)
24. Thomson BD, Marquez MD, Rawson S, Santos TMA, Harrison SC, Kahne DE. Structures of folding intermediates on BAM show diverse substrates fold by a uniform mechanism. *Proc Natl Acad Sci* **2026**; 123 (14): e2534936123. PMID: 41280068.
23. Santos TMA, Thomson BD, Marquez MD, Pan L, Monfared TH, Kahne DE. Native β -barrel substrates pass through two shared intermediates during folding on the BAM complex. *Proc Natl Acad Sci* **2024**; 121 (42): e2409672121. PMID: 39378083.

Publications Prior to Independent Career at Northeastern University

22. Ortiz B, Jennings J, Gross W, Santos TMA, Lin TY, Weibel DB, Lynn D. Soft materials that intercept, respond to, and sequester bacterial siderophores. *Chem Mater* **2021**; 33(13):5401–5412. PMID: 35341019.
21. Lee J, Tomasek D^{*}, Santos TMA^{*}, May MD^{*}, Meuskens I, Kahne D. Formation of a β -barrel membrane protein is catalyzed by the luminal surface of the assembly machine protein BamA. *Elife* **2019**; 8:e49787. PMID: 31724945.
20. Santos TMA, Lammers MG, Zhou M, Sparks IA, Rajendran M, Fang D, De Jesus CLY, Carneiro GFR, Cui Q, Weibel DB. Small molecule chelators reveal that iron starvation inhibits late stages of bacterial cytokinesis. *ACS Chem Biol* **2018**; 13(1):235–246. PMID: 29227619.
19. Hurley KA^{*}, Santos TMA^{*}, Fensterwald MR, Rajendran M, Moore JT, Balmond EI, Blahnik BJ, Faulkner KC, Foss MH, Heinrich VA, Lammers MG, Moore LC, Reynolds GD, Shearn-Nance GP, Stearns BA, Yao ZW, Shaw JT, Weibel DB. Targeting quinolone- and aminocoumarin-resistant bacteria with new gyramide analogs that inhibit DNA gyrase. *MedChemComm* **2017**; 8(5):942–951. PMID: 30034678.
18. Hurley, KA^{*}, Santos TMA^{*}, Nepomuceno GM, Huynh V, Shaw JT, Weibel DB. Targeting the bacterial division protein FtsZ. *J Med Chem* **2016**; 59(15):6975–6998. PMID: 26756351.
17. Lin TY, Santos TMA, Kontur W, Donohue TJ, Weibel DB. A cardiolipin-deficient mutant of *Rhodobacter sphaeroides* has an altered cell shape that impairs biofilm formation. *J Bacteriol* **2015**; 197(21):3446–3455. PMID: 26283770.

Highlighted as “Article of Significant Interest Selected by the Editors.”

16. Yin N, Stilwell MD, Santos TMA, Wang H, Weibel DB. Agarose particle-templated porous bacterial cellulose and its application in cartilage growth in vitro. *Acta Biomater* **2015**; 12:129–138. PMID: 25449918.

15. Foditsch C, Santos TMA, Teixeira AGV, Pereira RVV, Dias JM, Gaeta N, Bicalho RC. Isolation and characterization of *Faecalibacterium prausnitzii* from calves and piglets. *PLoS ONE* **2014**; 9(12):e116465. PMID: 25551453.
14. Santos TMA, Lin TY, Rajendran M, Anderson SM, Weibel DB. Polar localization of *Escherichia coli* chemoreceptors requires an intact Tol-Pal complex. *Mol Microbiol* **2014**; 92(5):985–1004. PMID: 24720726.

Highlighted by Faculty of 1000.

13. Yin N, Santos TMA, Auer GK, Crooks JA, Oliver PM, Weibel DB. Bacterial cellulose as a substrate for bacterial cell culture. *Appl Environ Microbiol* **2014**; 80(6):1926–1932. PMID: 24441155.
12. Machado VS, Bicalho ML, Meira-Junior EB, Rossi R, Ribeiro BL, Lima S, Santos TMA, Kussler A, Foditsch C, Ganda EK, Oikonomou G, Cheong SH, Gilbert RO, Bicalho RC. Subcutaneous immunization with inactivated bacterial components and purified protein of *Escherichia coli*, *Fusobacterium necrophorum* and *Trueperella pyogenes* prevents puerperal metritis in Holstein dairy cows. *PLoS ONE* **2014**; 9(3):e91734. PMID: 24638139.
11. Oliveira MNV*, Santos TMA*, Vale HMM, Cordero AFP, Tótola MR, Costa MD, Ferreira AB, Delvaux JC, Moraes CA, Borges AC. Endophytic microbial diversity in coffee cherries of *Coffea arabica* from Southeastern Brazil. *Can J Microbiol* **2013**; 59(4):221–230. PMID: 23586745.
10. Santos TMA, Bicalho RC. Diversity and succession of bacterial communities in uteri of postpartum metritic, endometritic and healthy dairy cows. *PLoS ONE* **2012**; 7(12):e53048. PMID: 23300859.

Highlighted Among the Top 10% Most Cited PLoS ONE Articles.

9. Santos TMA, Pereira RV, Caixeta LS, Guard CL, Bicalho RC. Microbial diversity in bovine papillomatous digital dermatitis in Holstein dairy cows from upstate New York. *FEMS Microbiol Ecol* **2012**; 79(2):518–529. PMID: 22093037.
8. Bicalho MLS, Machado VS, Nydam DV, Santos TMA, Bicalho RC. Evaluation of oral administration of bacteriophages to neonatal calves: Phage survival and impact on fecal *Escherichia coli*. *Livest Sci* **2012**; 144(3):294–299. PMID: 32362952.
7. Santos TMA, Ledbetter EC, Caixeta LS, Bicalho MLS, Bicalho RC. Isolation and characterization of two bacteriophages with strong in vitro antimicrobial activity against *Pseudomonas aeruginosa* isolated from canine ocular infection. *Am J Vet Res* **2011**; 72(8):1079–1086. PMID: 21801066.
6. Pereira RVV, Santos TMA, Bicalho ML, Caixeta LS, Machado VS, Bicalho RC. Antimicrobial resistance and prevalence of virulence factor genes in fecal *Escherichia coli* of Holstein calves fed milk with and without antimicrobials. *J Dairy Sci* **2011**; 94(9):4556–4565. PMID: 21854928.
5. Santos TMA, Gilbert RO, Bicalho RC. Metagenomic analysis of the uterine bacterial microbiota in healthy and metritic postpartum dairy cows. *J Dairy Sci* **2011**; 94(1):291–302. PMID: 21183039.
4. Santos TMA, Bicalho RC. Complete genome sequence of vB_EcoM_ECO1230-10: A coliphage with therapeutic potential for bovine metritis. *Vet Microbiol* **2011**; 148(2-4):267–275. PMID: 20875932.

3. Santos TMA, Caixeta LS, Machado VS, Rauf A, Gilbert RO, Bicalho RC. Antimicrobial resistance and presence of virulence factor genes in *Arcanobacterium pyogenes* isolated from uterus of postpartum dairy cows. *Vet Microbiol* **2010**; 145(1-2):84–89. PMID: 20346602.
2. Santos TMA, Gilbert RO, Caixeta LS, Machado VS, Teixeira LM, Bicalho RC. Susceptibility of *Escherichia coli* isolated from uteri of postpartum dairy cows to antibiotic and environmental bacteriophages. Part II: In vitro antimicrobial activity evaluation of a bacteriophage cocktail and several antibiotics. *J Dairy Sci* **2010**; 93(1):105–114. PMID: 20059909.
1. Bicalho RC, Santos TMA, Gilbert RO, Caixeta LS, Teixeira LM, Bicalho MLS, Machado VS. Susceptibility of *Escherichia coli* isolated from uteri of postpartum dairy cows to antibiotic and environmental bacteriophages. Part I: Isolation and lytic activity estimation of bacteriophages. *J Dairy Sci* **2010**; 93(1):93–104. PMID: 20059908.

HONORS

Faculty ACCESS Program	2026
Hanna H. Gray Faculty Award (Howard Hughes Medical Institute, HHMI)	2026
Hanna H. Gray Faculty Award (HHMI)	2024
Hanna H. Gray Postdoctoral Fellowship (HHMI)	2018
Academic Achievement Award (UW-Madison)	2017
Louis and Elsa Thomsen Wisconsin Distinguished Fellowship (UW-Madison)	2016
Burroughs Wellcome Fund Award (MBL) [Gratefully declined]	2014
Benjamin Kaminer Scholarship & Arthur Klorfein Scholarship and Fellowship (MBL)	2014
CNPq Fellowship for Graduate Research (UFV)	2006
President Arthur Bernardes Medal (UFV) [Equivalent of <i>summa cum laude</i>]	2006
Undergraduate Honor (UFV)	2006
FAPEMIG Fellowship for Undergraduate Research (UFV)	2005

INVITED LECTURES

17. Department of Chemistry, West Virginia University, Morgantown, WV (<i>Scheduled</i>)	2027
16. 1 st Brazilian Congress on Microbial Diversity, Brazil (<i>Scheduled</i>)	2026
15. 1 st FoodMicro Latino, 33 ^o Congresso Brasileiro de Microbiologia, Brazil	2025
14. VII Semana de Biologia, IFSULDEMINAS, Brazil (<i>Remote</i>)	2025
13. Department of Biology, Northeastern University, Boston, MA	2025
12. Empresa Brasileira de Pesquisa Agropecuária (Embrapa), Brazil	2024
17. Department of Pharmacological & Biomolecular Sciences, University of Milan, Italy	2024
10. Winter School, Universidade de São Paulo, Brazil	2024
9. VI Simpósio Internacional de Microbiologia and Biotecnologia, Brazil	2023
8. HHMI Hanna H. Gray Fellows Retreat, Janelia Research Campus, Ashburn, VA	2023
7. Department of Microbiology, Cornell University, Ithaca, NY	2023
6. Department of Biochemistry, University of Colorado-Boulder, Boulder, CO	2023
5. Department of Chemistry & Chemical Biology, Northeastern University, Boston, MA	2023

4. Department of Chemistry, Massachusetts Institute of Technology, Cambridge, MA 2023
3. Biophysics Seminar Series, University of Graz, Austria (*Remote*) 2022
2. HHMI Hanna H. Gray Fellows Retreat (*Remote*) 2022
1. ASM Microbe, Boston, MA (***Outstanding Student Abstract***) 2016

POSTER PRESENTATIONS

19. HHMI Science Meeting, Chevy Chase, MD 2025
18. HHMI Science Meeting, Chevy Chase, MD 2024
17. HHMI Science Meeting, Chevy Chase, MD 2023
16. HHMI Science Meeting, Chevy Chase, MD 2022
15. HHMI Science Meeting (*Remote*) 2022
14. HHMI Science Meeting (*Remote*) 2021
13. HHMI Science Meeting, Chevy Chase, MD 2019
12. HHMI Science Meeting, Ashburn, VA, USA 2018
11. 61th Annual Meeting of the Biophysical Society, New Orleans, LA 2017
10. Gordon Research Conference, New Antibacterial Discovery & Development, Italy 2016
9. The Molecular Genetics of Bacteria and Phages Conference, Madison, WI (***Poster Recognition Award***) 2015
8. 115th ASM General Meeting, New Orleans, LA (***Outstanding Student Poster***) 2015
7. 2015 Perlman Symposium on Antibiotic Discovery and Development, Madison, WI (***Perlman Symposium Poster Award***) 2015
6. The Molecular Genetics of Bacteria and Phages Conference, Madison, WI 2014
5. 20th Annual Boston Bacterial Meeting, Boston, MA 2014
4. 114th ASM General Meeting, Boston, MA (***Outstanding Student Poster***) 2014
3. 111th ASM General Meeting, New Orleans, LA 2011
2. 110th ASM General Meeting, San Diego, CA 2010
1. 109th ASM General Meeting, Philadelphia, PA 2009

TEACHING

Course Number	Course Title	Year	Number of Students	TRACE Mean Effectiveness (Max. = 5.0)
BIOL 4971	Jr/Sr Honors Project 2	Spring 2026	1	N/A
CHEM 5620	Protein Chemistry	Spring 2026	20	5.0
BIOC 4971	Jr/Sr Honors Project 2	Fall 2025	1	N/A
BIOL 4970	Jr/Sr Honors Project 1	Fall 2025	1	N/A
BIOC 4991	Research	Spring 2025	1	N/A
CHEM 5620	Protein Chemistry	Fall 2024	26	5.0
BIOL 4991	Research	Summer 2024	1	N/A
CHEM 5620	Protein Chemistry	Spring 2024	34	4.9

Additionally, I regularly serve as an instructor for:

- Dissertation Courses: CHEM 9990 (Dissertation Term 1), CHEM 9991 (Dissertation Term 2), CHEM 9996 (Dissertation Continuation)
- Seminar Courses: CHEM 5904 (Seminar), CHEM 8504 (Graduate Seminar)

RESEARCH TRAINEES

Current Group Members: 13

Advisee	Training Capacity	Period
Pandarinadh Chadarajupalli	Master's student	2026–present
Santhiyaa Ramaraj	Master's student	2026–present
Angela Tan	Master's student	2026–present
Pankti Lakhani	Master's student	2026–present
Pooshana Dhanasekaran	Master's student	2026–present
Yachne Elysee	Undergraduate student	2026–present
Ben Loew	Undergraduate student	2025–present
<i>- Recipient of Fall 2025 PEAK Award (2025)</i>		
Amy Ballesteros	Ph.D. student (G2)	2025–present
Thiago Sousa	Postdoctoral researcher	2025–present
<i>- Recipient of The American Institute of Chemists Award (2026)</i>		
Luiza Monteiro	Laboratory technician	2024–present
Arezou Dilmaghani-Marand	Ph.D. student (G3)	2024–present
Liana Annable	Ph.D. student (G3)	2024–present
Bakar Abbasi	Ph.D. student (G3)	2024–present

Former Group Members: 15

Advisee	Training Capacity	Period	Current Position
Gleucinei Castro	Visiting Ph.D. student (UEA, Brazil)	2025–26	<i>Anticipated graduation 2027</i>
Michael Arias	High school student (BC High)	2025–26	<i>Anticipated graduation 2026</i>
Michael Liotta	Undergraduate student	2026	<i>Applying for jobs</i>
Riley Raddohl	Undergraduate student	2025–26	<i>Anticipated graduation 2026</i>
Amalia Costa	Undergraduate student	2024–26	<i>Anticipated graduation 2027</i>
<i>- Recipient of the 2025–2026 Advancing Women in Science scholarship (2025)</i>			
Lucy Shaffer	Undergraduate student	2025–26	<i>Anticipated graduation 2028</i>
Anya Duarte	Undergraduate student	2025–26	<i>Anticipated graduation 2028</i>
Vignesh Vijayakumar	Master's student	2024–25	<i>Applying for jobs</i>
Atticus Conn	Undergraduate student	2024–25	<i>Applying to medical school</i>
Dylan Sacks	Undergraduate student	2024–25	<i>Postbaccalaureate Research Fellow, NIH</i>
Hannah Xu	Undergraduate student	2024–25	<i>Anticipated graduation 2028</i>
Alyssa Shemtov	Undergraduate student	2024–25	<i>Anticipated graduation 2026</i>

Grace Giarrizzo	Undergraduate student	2024–25	<i>Anticipated graduation 2026</i>
Caitlin Sia	Undergraduate student	2024–25	<i>Applying to medical school</i>
Andrea Harris	Undergraduate student	2024	Graduate student Life University

PROFESSIONAL MEMBERSHIP

The Protein Society	2026–present
American Association for the Advancement of Science	2022–present
American Chemical Society	2022–present
American Society for Microbiology	2009–present